

1.2.1 INCLUSIONS AND EXCLUSIONS

PARTICULAR	INCLUSION	EXCLUSION
Service	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing services, including sample preparation, sequencing and library preparation, and data analysis.	NA
Type	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing service types, such as amplicon sequencing, whole-genome sequencing, shotgun sequencing, and de novo sequencing.	NA
Technology	Qualitative and quantitative analysis (in terms of USD million) of technologies, such as sequencing by synthesis (SBS), sequencing by ligation (SBL), nanopore sequencing, and semiconductor sequencing.	NA
End User	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing service end users such as pharmaceutical and biotechnology companies, academic and research institutes, small-scale CROs, and CDMOs.	NA

1.3 MARKET SCOPE

1.3.1 MARKETS COVERED

MICROBIOME SEQUENCING SERVICES MARKET			
	SERVICE - Sample Preparation - Sequencing and Library Preparation - Data Analysis		TYPE - Amplicon Sequencing - Whole-genome Sequencing - Shotgun Sequencing - Other Types
			TECHNOLOGY - Sequencing by Synthesis - Sequencing by Ligation - Nanopore Sequencing - Other Technologies
		END USER	REGION
		- Pharmaceutical and Biotechnology Companies - Academic and Research Institutes - Other End Users	- North America (US and Canada) - Europe (Germany, UK, France, Italy, Spain, and Rest of Europe) - Asia Pacific (Japan, China, India, and Rest of Asia Pacific) - Latin America (Brazil and Rest of Latin America) - Middle East and Africa

Note 1: Other types of microbiome sequencing services include de novo sequencing (whole-genome sequencing without any reference genome), metagenomic sequencing, and metatranscriptomic sequencing.

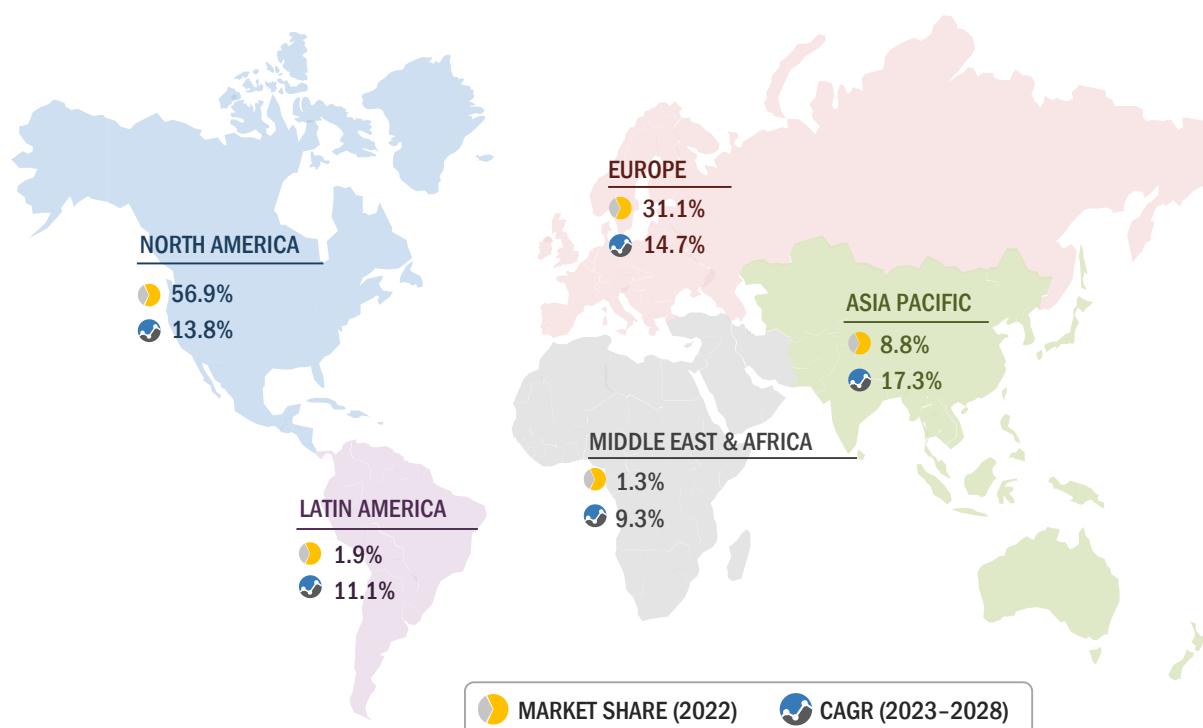
Note 2: Other microbiome sequencing technologies include semiconductor sequencing and Sanger sequencing.

Note 3: Other end users include hospitals and clinics, small-scale CROs, and CDMOs.

Note 4: The Rest of Europe includes Switzerland, Greece, the Netherlands, Belgium, Denmark, and Ireland.

Note 5: The Rest of Asia Pacific includes South Korea, Australia, and countries in Southeast Asia.

Note 6: The Rest of Latin America includes Argentina, Peru, Chile, and Colombia.

FIGURE 14 GEOGRAPHICAL SNAPSHOT OF MICROBIOME SEQUENCING SERVICES MARKET

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

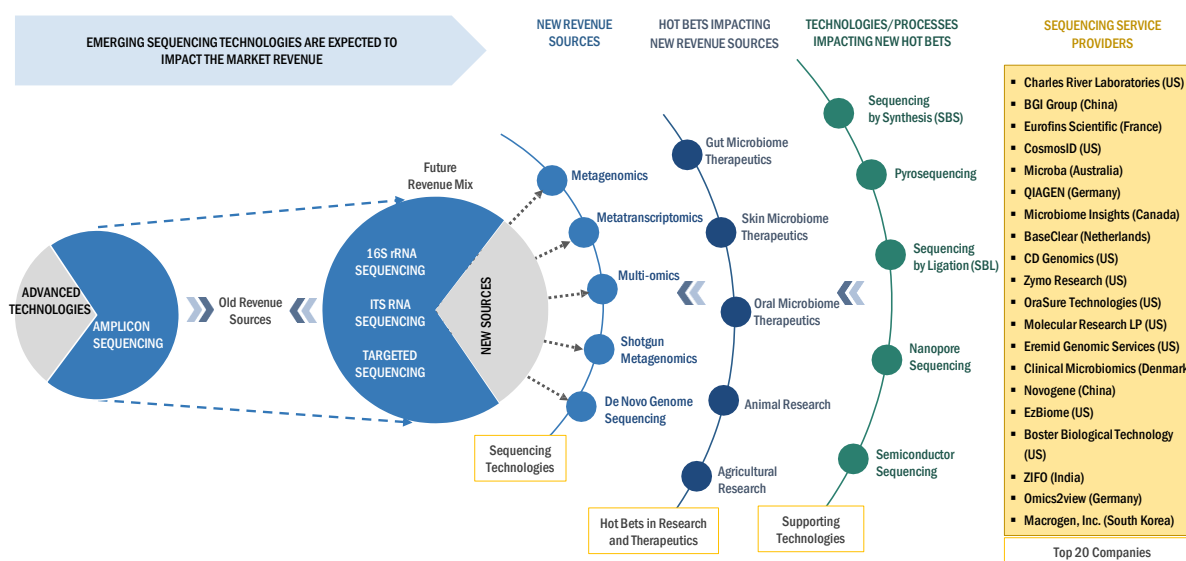
The microbiome sequencing services market is segmented into North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. Factors such as established research infrastructure, technological advancements, supportive regulatory environment, robust biotechnology and healthcare industry, and the availability of funding and investment are driving the growth of the microbiome sequencing services market in North America. The microbiome sequencing services market in the Asia Pacific is expected to register the highest growth rate during the forecast period due to the increasing awareness of microbiome health, the emerging healthcare and biotechnology industry, government support, and funding potential for personalized medicine.

The prominent players in the global microbiome sequencing services market are Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and Macrogen, Inc. (South Korea).

5.4 TRENDS AND DISRUPTIONS IMPACTING CUSTOMERS' BUSINESSES

Robust research activities to explore the potential of the human microbiome present great potential in a number of treatments against obesity, inflammatory bowel disease, and diabetes, among other conditions. Information on microbiomes helps in the identification of the causal links between the microbiome and disease, further facilitating the development of efficient and targeted treatment strategies. Additionally, progress in state-of-the-art sequencing approaches, computational analyses, and experimental assays is supporting the growing demand for microbiome sequencing.

FIGURE 20 REVENUE SHIFT AND NEW REVENUE POCKETS FOR MICROBIOME SEQUENCING SERVICE PROVIDERS



Source: MarketsandMarkets analysis

The most commonly used sequencing technologies include 16S rRNA sequencing, ITS RNA sequencing, and whole-genome sequencing. Sequencing and experimental analyses of the microbiome continue to advance substantially. By coupling multi-omics technologies, statistical and computational analyses, and more advanced disease models, these approaches promisingly provide the opportunity to establish disease causality and subsequently inform therapeutic development.

5.8.6 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

TABLE 10 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS: GENOMICS, MICROBIOME RESEARCH, AND METAGENOMICS

REGION	COUNTRY	REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS
North America	US	- National Institute of Standards and Technology (NIST) - National Human Genome Research Institute (NHGRI) - Joint Genome Institute (JGI)
	Canada	- The Genomics Research and Development Initiative (GRDI) - Genome Canada
Europe	Germany	- DZIF German Centre for Infection Research - HELMHOLTZ-ZENTRUM FÜR INFEKTIONSFORSCHUNG GMBH
	UK	- UK Research and Innovation (UKRI) - Microbiology Society
	France	- Paris Center for Microbiome Medicine (PaCeMM) FHU - Microbiome Foundation France
	Italy	- Microbiome Research Hub Italy - Council for Agricultural Research and Economics (CREA)
	Spain	- The Centre for Genomic Regulation (CRG) - National Centre for Genomic Analysis (CNAG-CRG) - The Higher Council for Scientific Research - The Institute for Research in Biomedicine (IRB)
Asia	Japan	- Human MetaGenome Consortium Japan (HMGJ) - Japan Agency for Medical Research and Development
	China	- Agricultural Genomics Institute at Shenzhen (AGIS) - National Genomics Data Centre China
	India	- Department of Biotechnology (GENOME India and Microbiome) - International Life Sciences Institute India (ILSI)
	South Korea	- National Research Foundation of Korea (NRF) - Korea Biotechnology Industry Organization (KoreaBIO)
	Australia	- National Research Infrastructure for Australia (Australian Microbiome Initiative) - Microbiome Research Centre (MRC)
Latin America	Brazil	- Brazilian Society of Genetics (Sociedade Brasileira de Genética in Portuguese) - The São Paulo Research Foundation (Fundação de Amparo à Pesquisa do Estado de São Paulo in Portuguese)

TABLE 146 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	0.21	0.25	0.29	0.34	0.40	0.46	0.55	0.64	17.3%
Sequencing by Ligation	0.06	0.07	0.08	0.10	0.11	0.13	0.15	0.18	16.8%
Nanopore Sequencing	0.02	0.03	0.04	0.04	0.05	0.07	0.08	0.10	22.2%
Other Technologies	0.17	0.20	0.23	0.27	0.31	0.36	0.42	0.49	16.0%
Total	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 147 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.15	0.18	0.21	0.25	0.29	0.35	0.41	0.49	18.7%
Academic and Research Institutes	0.21	0.25	0.29	0.33	0.39	0.45	0.53	0.62	16.5%
Other End Users	0.11	0.13	0.15	0.17	0.19	0.23	0.26	0.30	15.8%
Total	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.4 REST OF ASIA PACIFIC

The RoAPAC includes Australia, Taiwan, New Zealand, Thailand, South Korea, Singapore, and Malaysia, among other Asia Pacific countries. Factors such as public and private initiatives to support the adoption and development of microbiome technologies, increasing R&D expenditure, and increasing awareness of the diagnostic applications of microbiome sequencing are driving the growth of the RoAPAC market.

In September 2019, the South Korean government proposed plans to increase the budget of government subsidies in the biotech sector by 16% and finance a project to secure the genome data of 1 million people. The government aims to increase its budget spending to help the pharmaceutical sector develop innovative new drugs and medical devices. As per this agenda, the government will invest USD 951.8 million in R&D projects in the sector in 2020, with an increase of 16% in 2019. Over the past few years, a few events focusing on spreading awareness about the benefits of microbiome sequencing and the latest advancements in this field have been organized in the RoAPAC region.

In June 2022, Singapore-based precision gut microbiome company AMILI received USD 10.5 million in a Series A funding round led by technology and life sciences investor Vulcan Capital. New investors also joined the round, including Pruksha Group, TVM Capital Healthcare, Emtex Group, Capital Code, Pureland Group, Blue7, and Enterprise Singapore's investment arm SEEDS Capital.

11 COMPETITIVE LANDSCAPE

11.1 OVERVIEW

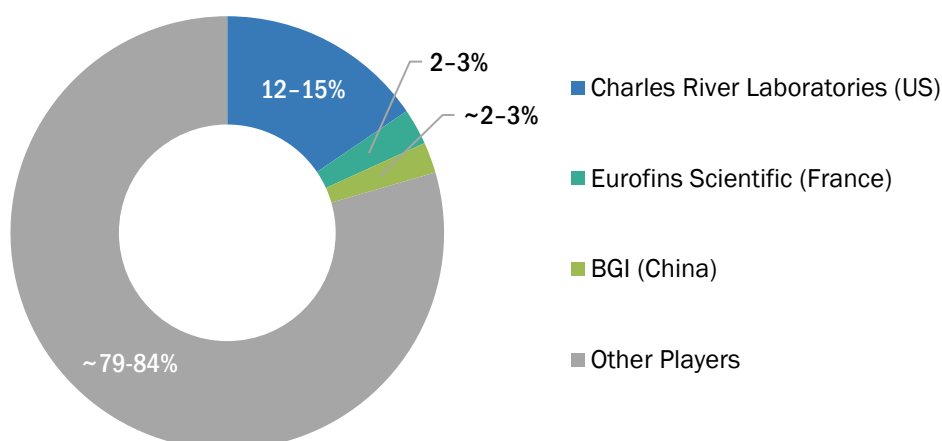
The competitive landscape includes an analysis of the key growth strategies adopted by major players between October 2020 and April 2023. Players in the microbiome sequencing services market have employed various strategies to expand their footprints and increase their market shares. The key strategies adopted by major players in the microbiome sequencing services market include expansions, service launches, acquisitions, agreements, and collaborations.

Prominent players in this market include Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and Macrogen, Inc. (South Korea).

11.3 MARKET SHARE ANALYSIS

The global microbiome sequencing services market is a fragmented market, with Charles River Laboratories (US), Eurofins Scientific (France), and BGI (China) together accounting for ~20% of the global market. Players such as CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and MacroGen, Inc. (South Korea) accounted for the remaining 75–80% of the total market.

FIGURE 29 MICROBIOME SEQUENCING SERVICES MARKET SHARE ANALYSIS, BY KEY PLAYER, 2022



Other players include CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and MacroGen, Inc. (South Korea).

Source: Annual Reports, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 169 MICROBIOME SEQUENCING SERVICES MARKET: DEGREE OF COMPETITION

DEGREE OF COMPETITION	FRAGMENTED
Total Market Share of the Top 3 Players	~20%
Other Players	~80%
THE DEGREE OF COMPETITION IS DEFINED AS BELOW:	
Fragmented: When the top 5 players have a total market share of <25%	
Competitive: When the top 5 players have a total market share of 25–50%	
Consolidated: When the top 5 players have a total market share of >50%	

Note: Market shares of the above companies are calculated based on relevant segmental revenues generated by the companies.
Source: Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets

11.5 COMPANY EVALUATION QUADRANT

The company evaluation quadrant provides information on the top 20 players offering microbiome sequencing services and outlines findings on the performance of each vendor. The industry matrix evaluations are based on two broad categories—Service Offerings and Business Strategies. Each type carries various criteria based on which the companies have been evaluated. The evaluation criteria considered under service offerings include the breadth of offerings, services launched, service technologies, and support. The evaluation criteria considered under the business strategy includes reach (based on market and geographic presence), industry coverage (based on end users catered to), viability, and inorganic growth strategies. The top 20 companies are placed into four categories based on their performance in each criterion—“Stars,” “Emerging Leaders,” “Pervasive Players,” and “Participants.”

We have evaluated 20 companies in this section, which include Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and MacroGen, Inc. (South Korea).

11.5.1 STARS

Players who fall into this category receive high scores for most evaluation criteria. They have a strong and established product portfolio, high market shares, comprehensive application coverage, and a strong geographical footprint. These players provide mature, innovative, and reputable microbiome sequencing services. Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), and QIAGEN (Germany) are the star players in this market.

11.5.2 EMERGING LEADERS

Emerging leaders have demonstrated substantial product innovations compared to their competitors (resulting in an evolved service portfolio) and significant market presence. However, they do not have robust growth strategies for geographical penetration and overall business growth. Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), and OraSure Technologies (US) are emerging leaders in the market.

11.5.3 PERVASIVE PLAYERS

Companies falling in this quadrant are established players with very strong business strategies and global distribution/product availability. However, they have a weaker service portfolio and lower market share than the emerging leaders. Pervasive players are Clinical-Microbiomics A/S (Denmark) and Novogene Co., Ltd. (China).

11.5.4 PARTICIPANTS

The companies that fall in this category have niche service offerings. They do not have substantial business strategies as compared to other established vendors. These companies could be new entrants and require more time before getting traction. Participants in the microbiome sequencing services market include MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and MacroGen, Inc. (South Korea).

FIGURE 31 MICROBIOME SEQUENCING SERVICES MARKET: COMPANY EVALUATION QUADRANT, 2022



Source: MarketsandMarkets Analysis

12.2.5 MACROGEN, INC.

MACROGEN, INC.

Year Founded	1997
Headquarters - Country	Republic of Korea
Headquarters – City/State	Seoul
Ownership	Private
Business overview	The company offers personalized healthcare solutions by analyzing each person's gut microbiome based on big data.
Product	Whole-genome Sequencing and Metagenomic Sequencing
Geographical Presence	South Korea